

Stacking Signal TSV for Thermal Dissipation in Global Routing for 3-D IC

Po-Yang Hsu, Hsien-Te Chen, and TingTing Hwang

Abstract—With no further shrinking of device size, 3-D chip stacking by through-silicon-via (TSV) has been identified as an effective way to achieve better performance in speed and power. However, such solution inevitably encounters challenges in thermal dissipation since stacked dies generate a significant amount of heat per unit volume. We leverage an integrated design methodology of stacked-signal-TSVs to minimize temperature. Based on this structure, a three-stage TSV locating algorithm in global routing is designed. We demonstrate that our results, compared with baseline circuits, have 17% temperature reduction with 3% wiring overhead and no performance loss calculated by 3-D Elmore delay model. Compared with the paper by Cong and Zhang where additional thermal TSVs are inserted, our experimental results have in average 23% less TSVs with the same temperature constraint. Compared with the paper by Pathak and Lim, where movable signal TSVs are relocated to reduce temperature in hotspot regions, our result has 8% more temperature reduction with the same number of signal TSVs.

Index Terms—Global routing, stacked signal TSV, thermal dissipation, 3-D IC, through-silicon-via (TSV).

I. INTRODUCTION

WITH NO FURTHER shrinking of device size, 3-D chip stacking by through-silicon-via (TSV) has been identified as an effective way to decrease the wirelength of circuit and achieve better performance in speed and power [4], [5]. However, such solution inevitably encounters challenges in thermal dissipation since stacked dies generate significant amount of heat per unit volume, and the IC stacking styles also restrict the thermal dissipation ability due to the heat flow blocked by insulation (bonding) layers.

To cope with the thermal dissipation issue, a lot of previous paper has been proposed [2], [6]–[10]. Cong and Zhang [2] proposed a thermal TSV insertion method to add extra thermal vias to lower the thermal resistance. Given a temperature constraint, their algorithm computes the location of thermal TSV taking into consideration power density of circuit. Zhang *et al.* [11] proposed a temperature-aware 3-D global routing algorithm by inserting both additional thermal vias and thermal wires to achieve better thermal dissipation

of hotspot area. Onakaraiah and Tan [12] further discussed the chip temperature impact by adding thermal TSV under different thermal TSV settings and 3-D stacking styles.

Nevertheless, TSVs are usually ten times larger than logic gates and memory cells [1]. Therefore, these thermal TSV insertion methods reduce the temperature at the cost of extra area of thermal vias and thermal wires [2], [7]–[10]. To avoid additional thermal TSV overhead, Pathak and Lim [3] proposed a performance driven and thermal-aware Steiner routing algorithm to place signal TSV to reduce temperature without additional thermal TSV and performance loss. In their approach, signal TSVs are classified to movable and nonmovable TSVs by 3-D Elmore delay model. The movable signal TSVs are those signal TSVs whose movement will not degrade performance and are placed near hotspot area. However, this method does not utilize the outstanding thermal conductance of TSV in thermal dissipation. That is, stacked TSVs provide a direct thermal conducting path from a heat source to a heat sink in vertical fashion. To fully utilize TSV thermal conductance, Chen *et al.* [13] proposed a stacked-TSV power network structure to improve thermal dissipation in power network. The heat flow can be conducted to the heat sink effectively by stacked TSVs.

In this paper, we propose a three-stage TSV locating algorithm in global routing for 3-D IC to minimize temperature with small wiring overhead. The principle of our algorithm is to construct stacked signal TSV and utilize the thermal conductance of stacked TSVs to improve the thermal dissipation of the circuit. The structure of the stacked signal TSV in signal interconnections is indebted to the stacked TSV structure in the power network for thermal dissipation proposed by Chen *et al.* [13]. The contributions of our method are the following:

- 1) *Full Utilization of TSV's Thermal Conductance*: Stacked TSVs have greater thermal conductivity than non-stacked ones. To fully utilize the thermal dissipation capability of stacked TSVs, we develop the first algorithm to stack signal TSVs.
- 2) *No Extra TSV Area Overhead*: Without adding extra thermal TSVs, aligned signal TSVs are constructed to dissipate heat vertically. Unlike the thermal via insertion algorithm, our approach does not incur additional area overhead for TSVs.
- 3) *Small Wiring Overhead and No Performance Loss*: A gain function is proposed to simultaneously consider the interconnect wirelength and temperature profile. Our

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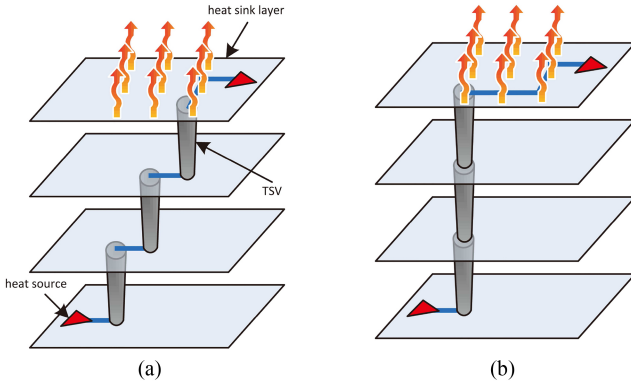


Fig. 1. Structure of nonstacked TSVs and stacked TSVs for signal wires. (a) Nonstacked signal TSVs. (b) Stacked signal TSVs.

algorithm can use this gain function to optimize the chip temperature with small wiring overhead. Moreover, by using the 3-D Elmore delay model [3] to get the timing information of circuit, our algorithm moves TSVs on non-critical paths. Hence, there incurs no performance loss.

The rest of the paper is organized as follows. Section II describes our motivation. Section III shows thermal impact of distance between stacked signal TSV and current source. In Section IV, a TSV locating algorithm to fully utilize stacked signal TSV is proposed. Finally, Section V shows experimental results, and Section VI concludes the paper.

II. MOTIVATION

The structure of stacked TSVs used in power network can be utilized to construct stacked signal TSV [13]. Conceptual structures of nonstacked TSVs and stacked TSVs for signal wires in 4-tier 3-D IC are illustrated in Fig. 1.

To understand the effectiveness of heat dissipation of these two structures, we conduct an experiment on these two structures at a given power density. The bonding style of TSV is assumed to be face-to-back approach. Only one signal running from the bottom tier to the top tier is tested where each tier has one TSV to connect the signal. The heat sink is placed at the topmost tier. The heat source is placed at the bottommost tier and connected to signal TSV with $20\mu\text{m}$ long metal wire (assumed that the width of metal layer is the same as the width of signal TSV, say $5\mu\text{m}$ in this analysis). For nonstacked signal TSV cases, the distances of TSVs between tiers are minimum distances of TSV pitch ($1\mu\text{m}$ to $10\mu\text{m}$ based on TSV size) and $20\mu\text{m}$ while for stacked signal TSV case, the signal TSVs are aligned at the same location.

The key parameters in this simplified analysis case are shown in Table I. In this simplified analysis, the thermal dissipation of metal layer is not taken into account because it is surrounded by dielectric material where the thermal resistance of low-K material ($10^6\mu\text{m} \cdot \text{K/W}$) is very high as compared to that of metal layer ($2500\mu\text{m} \cdot \text{K/W}$). The thermal dissipation of metal layer has no effect because the heat is blocked at low-K material.

TABLE I
KEY PARAMETERS

Parameter	Value
No. of tiers	4
Thickness of wafer	$50\mu\text{m}$
Area of TSV	$25\mu\text{m}^2$
Area of Die	$90000\mu\text{m}^2$
Thickness of SiO_2 in TSV	$0.2\mu\text{m}$
Thickness of dielectrics	$2\mu\text{m}$
Thermal resistance of silicon	$6670\mu\text{m} \cdot \text{K/W}$
Thermal resistance of copper	$2500\mu\text{m} \cdot \text{K/W}$
Thermal resistance of low-K material	$10^6\mu\text{m} \cdot \text{K/W}$
Power density of heat source	72.2 W/cm^2
Ambient temperature	25°C

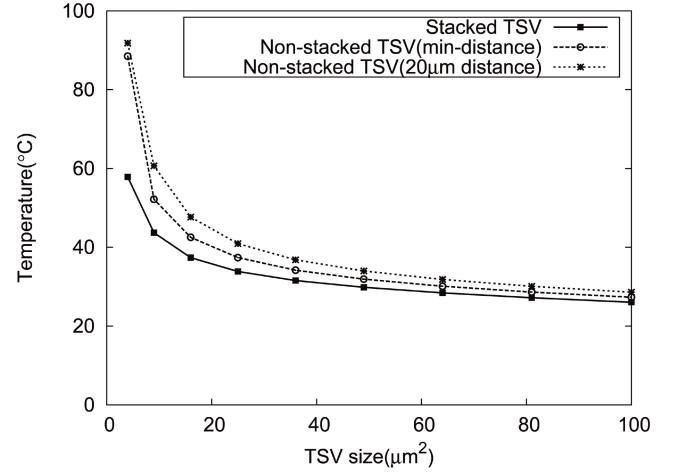


Fig. 2. Temperatures of stacked TSV, nonstacked TSV at minimum distance and at $20\mu\text{m}$ distance.

Fig. 2 illustrates comparative results of temperature for different sizes of TSV. The horizontal axis in Fig. 2 is the area of TSV while the vertical axis is the temperature. Compared with nonstacked TSV structures at minimum distance and $20\mu\text{m}$ distance, stacked signal TSV structure shows its significant effectiveness in temperature reduction for various sizes of TSV.

This example demonstrates that proper placement of signal TSVs is an effective approach to temperature reduction in 3-D IC and motivates us to place TSV so that there can be as many stacked signal TSVs as possible. Based on this observation, we will study location assignment of signal TSVs in global routing with the objective of minimizing temperature and wirelength.

III. LOCATION OF STACKED SIGNAL TSV

A. Thermal Model

To evaluate thermal dissipation, we adopt the resistive thermal model for simulation [2], [14]–[17]. This thermal model is similar to the traditional power network. An equivalent resistive thermal circuit through thermal-electrical analogy is constructed and thermal resistors are connected between nodes representing spatially adjacent regions. Finally, thermal current sources are mapped to power source.

As shown in Fig. 3, the compact thermal model divides the 3-D circuit into tile array composed of many grid nodes.

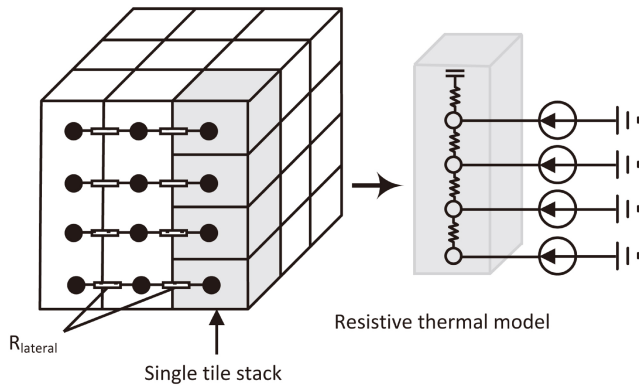


Fig. 3. Thermal model.

These grids are connected with lateral thermal resistor denoted by $R_{lateral}$. The lateral thermal resistors are determined by heat conductance of device material. Besides, in vertical direction, both device material and TSV affect the vertical thermal dissipation. Each single tile stack has its vertical thermal resistors as shown in Fig. 3. Thus, the heat flow can be conducted to the heat sink by lateral and vertical dissipation. In 3-D IC, the heat sink is usually located on the bottommost or topmost layer, and the boundary of other dies is adiabatic.

From the thermal current and the resistors, we can obtain the voltages that represent the temperatures of grids in this equivalent thermal circuit. In our thermal modeling, this grid-based method is applied for thermal simulation and analysis.

B. Distance of Stacked Signal TSV to Heat Source

In previous 3-D thermal-aware research, Cong and Zhang [2], [9] inserted thermal via estimating temperature at macro block level given power dissipation at macro block while Goplen and Sapatnekar [7], [8] inserted thermal via cell level given power dissipation of each cell. None of the previous paper except Pathak and Lim's [3] considered heat dissipation at netlist level. In this paper, we utilize the stacked signal TSV in a net to dissipate heat. First, the cell power is modeled as independent current source [18]. The results from this kind of approach are consistent with experimental results [19]. Therefore, the current source (represent the current consumed by cell) is equivalent to the heat source of a net in the compact thermal model as described by aforementioned [18].

To observe thermal dissipation with the stacked signal TSV, we perform the following experiment. Under the same simulation condition as the stacked signal TSV case in Table I, we conduct this experiment with various distances of source nodes of a net (equivalent to heat source) to stacked signal TSV. The result is shown in Fig. 4. The vertical axis in Fig. 4 is the temperature of the current (heat) source while the horizontal axis is the distance of the stacked signal TSV to the heat source. From this figure, we observe that the nearer to heat source the stacked signal TSV structure is, the more effective in thermal dissipation the stacked signal TSV structure performs. Therefore, to fully utilize stacked signal TSV structure in thermal dissipation, the distance of stacked signal TSV to heat source should be taken into account in allocation of signal TSVs.

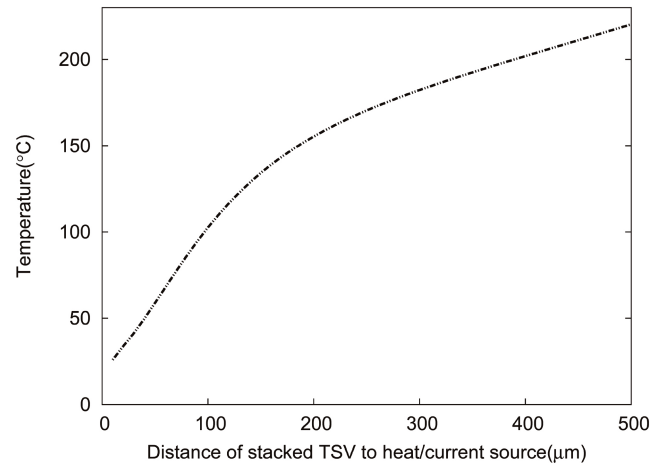


Fig. 4. Relationship between temperature and distance of stacked signal TSV to heat source.

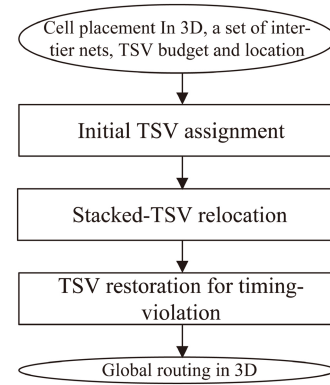


Fig. 5. Overall flow of placing signal TSVs in global routing.

IV. SIGNAL TSV ASSIGNMENT AND RELOCATION FOR THERMAL DISSIPATION

A. Overview of the Algorithm

Based on the above experiment, our goal is to develop an algorithm to stack signal TSVs close to hotspot regions so that the vertical heat dissipation path can be constructed without adding extra thermal TSV. The objective of our algorithm to place signal TSVs is to minimize the temperature and wiring overhead without any performance loss. Fig. 5 shows the flow of our algorithm. The inputs to the algorithm are a 3-D netlist file of placed cells, TSV budget, and their initial locations. The information of TSV budget includes the total number of TSVs and the number of TSVs for each two adjacent tiers. This TSV budget planning (TSV input location) is done during cell placement [1]. Then, the first stage is an initial TSV assignment stage. In this stage, we will assign signal TSVs to a suitable grid location by Steiner tree solution. With the guidance of Steiner tree, the goal is to have as many stacked TSVs as possible without increase in wirelength. The second stage is a stacked-TSV relocation stage. In this stage, the temperature reduction is the main consideration. The stacked signal TSVs will be relocated close to the hotspot grids to reduce temperature with minimum increase in wirelength. To avoid any performance loss, the timing of a net will be

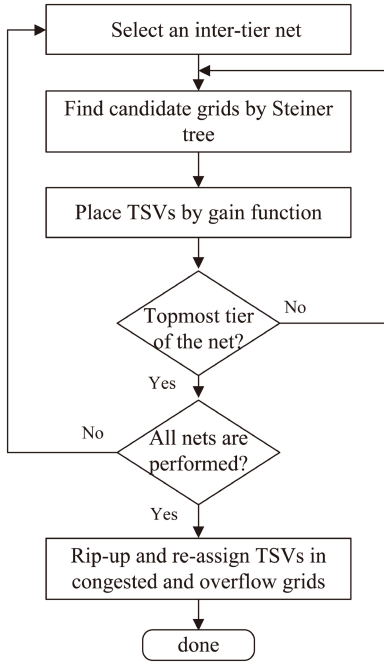


Fig. 6. Initial TSV assignment flow.

checked at step TSV restoration for timing violation stage. If the original timing is violated, some moved TSVs are selected to be restored to their initial locations from inputs by minimum set covering algorithm. Finally, the output is the placement of TSVs. The location of TSVs can be viewed as extra terminals in a net. Then, any global routing tool can be utilized to complete the global routing.

B. Initial TSV Assignment Stage

In this section, we will describe our initial TSV assignment algorithm. Fig. 6 shows the flow. Let the number of tiers be k and the tier is numbered from bottom to top. Intertier net is processed net by net based on its priority. The priority of a net is defined as the number of tiers that the net crosses. The larger the number is, the higher the priority is. Let one intertier net cross from tier l to tier k . Terminals belong to the bottom two tiers (tier l and tier $l+1$) are processed first. Then, terminals in the next two adjacent tiers (tiers $l+1$ and $l+2$) are processed. Terminals of intertier net are processed tier by tier from bottom to top until we reach the top tier k that the net crosses. For each two adjacent tiers, the step of *Find candidate grids by Steiner tree* is used to find all candidate grids to place TSVs, and the exact TSV locations are determined by the step of *Place TSVs by a gain function*. Note that after tiers l and $l+1$ are processed, TSVs connectivity between two tiers are determined. As 2-D Steiner routing used in our *Initial TSV assignment algorithm* is executed tier by tier, our method allows a better chance to form stacked TSVs as compared with a 3-D Steiner routing. In the following section, we will introduce the steps of *Find candidate grids by Steiner tree*, *Place TSVs by a gain function*, and *Rip-up and reassign TSVs in congested and overflow grids*.

1) *Find Candidate Grids by Steiner Tree*: In the step of *Find candidate grids by Steiner tree*, our goal is to find candidate location to place as many stacked TSVs as possible

without increase in wirelength. For an intertier net, the terminals between two adjacent tiers that the net crosses are mapped into the same 2-D plane, and the Steiner tree is constructed. Next, based on this Steiner tree, bounding boxes are built for two-pin segments that are intertiers. To connect sink nodes in upper tier and lower tier, the grids in these bounding boxes are candidate grids for planning TSV. Fig. 7 demonstrates an example of building bounding boxes for intertier nets. Fig. 7(a) shows terminals in two adjacent tiers where $\{A, B, C, D\}$ are at tier k and $\{X, Y, Z\}$ at tier $k-1$, Fig. 7(b) shows their Steiner tree for terminals mapped in 2-D plane, and Fig. 7(c) shows bounding boxes of intertier two-pin segments, $\{D, X\}$, $\{C, Y\}$ and $\{A, Z\}$. Grids in these bounding boxes are candidates for TSV assignment.

2) *Place TSVs by a Gain Function*: To connect sink nodes in upper tier and lower tier requires a TSV. If a TSV is assigned within the bounding box, then no extra wire incurs. Therefore, all grids in the bounding boxes are candidate grids for us to place TSVs. Nevertheless, to consider temperature reduction factor, it is preferable to place a TSV close to high temperature region. Therefore, in the step of *Place TSVs by a gain function*, for each candidate grid (i, j, k) in the bounding boxes, we define $Gain_{i,j,k}$ in (1) to consider temperature reduction factors

$$Gain_{i,j,k} = SD_{i,j} \times \frac{\sum_{k=1}^n PowDensity_{i,j,k}}{TSVNum_{i,j,k}}. \quad (1)$$

First, $PowDensity_{i,j,k}$ is the power density in grid (i, j, k) where i, j, k denote the coordinates of the grid node in x, y, z axis direction and n is the number of tiers in the design. In general, a high-lumped power density grid needs more signal TSVs to dissipate its heat. Second, the $TSVNum_{i,j,k}$ is the number of signal TSVs in grid (i, j, k) . Under the same power density, the thermal dissipation ability at grid (i, j, k) is inferior while $TSVNum_{i,j,k}$ is small. The heat flow cannot be delivered to the heat sink efficiently in vertical direction. Hence, it is more desirable to assign a TSV at a grid with small $TSVNum$.

Finally, we consider the stacking degree factor. The algorithm will attempt to stack signal TSVs so that the heat can be dissipated vertically. $SD_{i,j}$ is defined as the stacking degree in grid (i, j) , which is computed as the number of TSV stacking at grid position (i, j) . The more signal TSVs are placed below grid (i, j) , the more desirable for placing TSV at grid (i, j, k) . In other words, a larger $Gain$ value means higher power density, less TSVs, and more stacking signal TSVs.

After calculation of the gain functions for all grids in the bounding box formed by a candidate intertier two-pin segments, the grid with the largest gain value is selected for the bounding box and denoted as *highest_gain_grid*. *Highest_gain_grids* are selected for all bounding boxes. Note that one TSV is sufficient to connect terminals of a net crossing tiers. However, to consider shorter wirelength, more than one TSV may be inserted. Hence, the second step is considering the given TSV budget N , to select N *highest_gain_grids* with the largest gain values. Therefore, the total number of TSV assigned in the intertier net is determined by TSV budget, not by the number of bounding boxes.

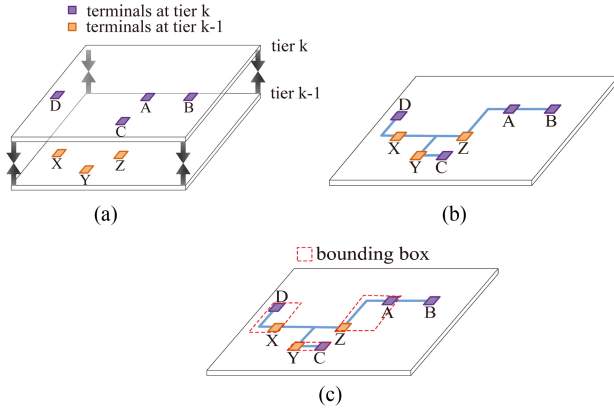


Fig. 7. Candidate grids within bounding boxes by bottom-up Steiner tree. (a) Terminals at tier k and $k-1$. (b) Steiner tree for terminals mapped in 2-D plane. (c) Bounding boxes for intertier two-pin segment.

3) *Rip-Up and Reassign TSVs in Congested and Overflow Grids*: the objective of this step is to rip-up and reassign TSVs in congested and overflow grids. To identify congested grids, we define an estimation function as follows. At first, the Steiner tree of a net is generated by FLUTE [20] and then all two-pin segments of the tree are routed by pattern route. For each grid of a global routing model, its routing demand is computed. Then, the grids with the highest routing demand will be specified as congested grids and TSVs in congested grids will be reassigned to low routing demand grids nearby. As to overflow grid, we define it as the grid whose whitespace is negative. Consequently, we reassign TSVs in overflow grids to grids nearby that have sufficient whitespace to satisfy the area constraint.

C. Stacked-TSV Relocation Stage

In our initial TSV assignment stage, TSVs are assigned to construct as many stacked TSVs as possible. Since wirelength overhead is our main consideration in this stage, only candidate grids within the bounding boxes are considered for TSV insertion. However, not all hotspots are near the candidate grids. When hotspots are outside and away from the bounding boxes of Steiner trees, the temperature cannot be reduced. For this reason, stacked-TSV relocation stage is proposed to solve this problem. The basic idea of stacked-TSV relocation is to move stacked signal-TSVs to hotspot locations with minimum wirelength increase. Note that after initial TSV assignment step, TSVs that are stacked, are determined. When we say stacked TSV relocation, we mean moving the whole stacked TSVs at the same 2-D location simultaneously.

Fig. 8 shows the flow of stacked-TSV relocation stage. First, starting from the bottommost tier to the topmost tier, TSV relocation is performed. In each tier i , the hotspot grids will be identified first. Then, from hotspot grids, a region is extended to explore their saver nets. Here, saver nets are defined as those nets across tier i and whose TSVs are candidates to be moved to hotspot grid. In one iteration, only one saver net is utilized to move its stacked TSV to one hotspot grid for heat dissipation. Then, in the step of *Selection of saver net and signal TSVs*, a flow network model is proposed to

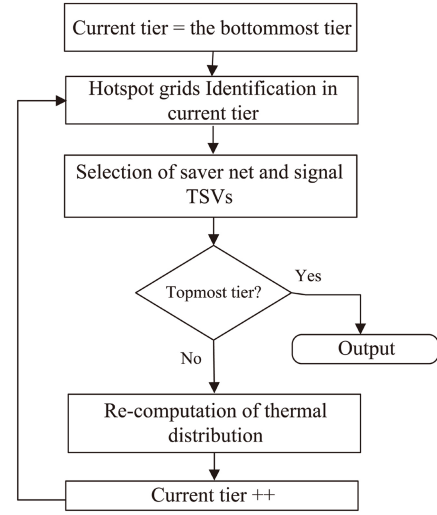


Fig. 8. Stacked-TSV relocation flow.

select the best saver net for each hotspot grid and the best TSVs in saver net to construct stacked TSV simultaneously. By using a minimum cost maximum flow algorithm, the best stacked TSVs of saver net will be relocated to hotspot grids. The stacked TSV can only be relocated once. The selected saver nets and its stacked TSVs will be locked in the following iterations. After tier i is processed, the thermal distribution of circuit will be recomputed. The next iteration continues till it reaches the topmost tier. In this TSV relocation stage, the temperature at hotspots is improved iteration by iteration. After one relocation step is completed, if the temperature is still higher than a given goal, the same relocation step can run again.

In our algorithm, the hotspot grids are identified by the top 10% highest thermal-critical grids. The thermal criticality of grid in (i, j, k) is defined by $ThermalCri_{i,j,k}$ as follows:

$$ThermalCri_{i,j,k} = \frac{\sum_{k=1}^n PowDensity_{i,j,k}}{TSVNum_{i,j,k}}. \quad (2)$$

The factors of this equation are similar to those in (1).

In the following section, we show how to solve selection of saver net and signal TSVs by a minimum cost network flow problem.

1) *Selection Issues for TSV Relocation*: First, for a hotspot grid, we define a region to find its saver net. The region is defined as a circle where the center of circle is the hotspot grid, and the radius is k grids long (In our experiment, k is set to 4). In this region, all nets are potential saver nets for the hotspot grids. For example, in Fig. 9, there are three hotspot grids, h_a , h_b , and h_c , and four saver nets, net s_1 , s_2 , s_3 , and s_4 . In the figure, the dotted line defines the search region of saver net for each hotspot grid with $k = 2$. For hotspot h_b , its saver nets are $\{s_2, s_4\}$ because nets s_2 and s_4 are in the 2-grid long region. Similarly, for hotspot grids h_a and h_c , their potential saver nets are $\{s_1\}$ and $\{s_3, s_4\}$, respectively.

Fig. 10(a) draws the selection issues of hotspot grids and saver net. As shown in this example, the stacked TSVs of

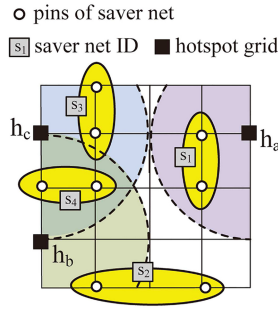


Fig. 9. Saver net exploration by search region ($k = 2$).

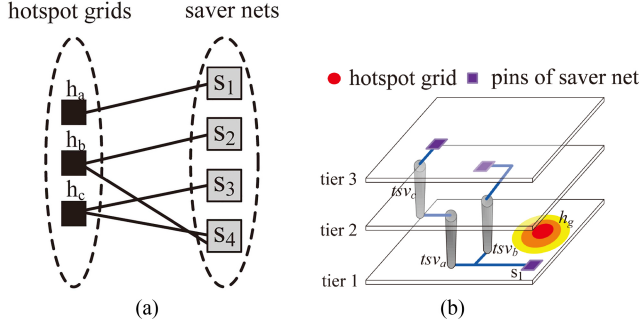


Fig. 10. Selection issues for TSV relocation. (a) Selection of saver nets for hotspot grids. (b) Selection of stacked TSVs.

a saver net is useful to dissipate the heat in many hotspots. However, only one hotspot location can be selected. For example, net s_4 is useful for grids h_b and h_c . However, the stacked TSVs of net s_4 can move to location near either grid h_b or grid h_c but not both. Since the selection of the best saver net for hotspot grids may conflict with each other, all hotspot grids should be taken into consideration at the same time.

After the matched saver net for each hotspot grid is determined, we are to answer the next selection problem as to which TSV in saver net is to be stacked in hotspot grids. The temperature reduction could vary according to different locations of stacked TSVs in the saver net. Fig. 10(b) illustrates an example. In Fig. 10(b), let s_1 be a saver net of hotspot grid h_g , and there are three TSVs, tsv_a, tsv_b, tsv_c in saver net s_1 . We are to move one TSV to grid h_g . In tier 1, either TSV tsv_a or TSV tsv_b can be moved to grid h_g while in tier 2, only TSV c can be moved. This figure shows that for each tier, there may be more than one TSV in saver net and we can move any one of them to the hotspot grid. However, suppose that the location of TSV tsv_a in saver net is also a hotspot grid. Then, the circuit temperature is not improved if we relocate TSV tsv_a to hotspot grid h_g . Therefore, the location of TSVs in saver nets should be taken into consideration while selecting TSVs to save their hotspot grid.

2) *Network-Flow Optimization for TSV Relocation*: To select the best saver net for each hotspot grid and find the best TSVs in the saver net to construct stacked TSV simultaneously, we model the above selections as a network flow optimization problem in the step of *Selection of saver net and signal TSVs*. The flow network is denoted by $G(V \cup E)$, where node set $V = \{s \cup H_g + S_n \cup T_1 + S_{tsv_1} \cup T_2 + S_{tsv_2} \cup T_3 +$

$\dots + S_{tsv_{k-1}} \cup T_k + t\}$ and edge set $E = \{IE \cup SE + CE \cup TE_1 + LCE_1 \cup TE_2 + LCE_2 \cup \dots \cup TE_{k-1} + LCE_{k-1} \cup OE\}$ where k is the number of tier in the design. In this flow network, s and t represent the source node and destination node of network flow, respectively, each node $h_i \in H_g$ corresponds to a hotspot grid, each node $s_i \in S_n$ corresponds to a saver net, each node $t_{s_i,j} \in T_j$ corresponds to *saver_net_at_tier* (a saver net s_i at tier j connecting tier $j+1$ by some TSV), and each node $tsv_{s_i,j,n} \in S_{tsv_j}$ corresponds to a signal TSV n in saver net s_i connecting tier j and tier $j+1$. IE, SE, CE, TE_j, LCE_j , and OE represent incoming edge, saver net assignment edge, connecting edge, TSV assignment edge, layer connecting edge, and outgoing edge, respectively. Incoming edges are constructed from source node s to all nodes in H_g , and outgoing edges are constructed from the topmost *saver_net_at_tier* nodes to destination node t . Connecting edges are constructed from a saver net node s_i in S_n to a *saver_net_at_tier* node $t_{s_i,1}$ in T_1 . Layer connecting edges LCE_j are constructed from a signal TSV $tsv_{s_i,j,n}$ in S_{tsv_j} to a *saver_net_at_tier* node $t_{s_i,j+1}$ in T_{j+1} .

There are two types of assignment edge, saver net assignment edges SE and TSV assignment edges TE_j . An edge in saver net assignment edge, e_{h_i,s_i} connecting a hotspot grid h_i in H_g and a saver net s_i in S_n , represents that the saver net s_i can be selected to move its stacked TSV for hotspot grid h_i . An edge in TSV assignment edges TE_j , $e_{t_{s_i,j},tsv_{s_i,j,n}}$ connecting a *saver_net_at_tier* node $t_{s_i,j}$ in T_j and a signal TSV $tsv_{s_i,j,n}$ in S_{tsv_j} , represents that signal TSV n linking tier j and $j+1$ in saver net s_i can be selected.

The capacity of all edges including incoming/outgoing/connecting/layer connecting edges, and assignment edges is set to 1. For saver net assignment edges, this capacity will ensure that each hotspot grid can find one and only one matched saver net to relocate stacked TSV for temperature reduction. For TSV assignment edges, this capacity will ensure that one and only one TSV in one tier is selected for stacking. As to the edge cost, for incoming/outgoing/connecting/layer connecting edges, they are all set to 0. As to saver net assignment edges and TSV assignment edges, we consider the factors of power density, TSV number, wirelength overhead, whitespace density, and stacking degree of saver net in the cost function. The details will be described in the following sections. After we model this flow network problem, a minimum cost maximum flow algorithm is used to find saver nets and their specific signal TSVs for stacking.

Let us give an example to demonstrate the modeling of flow network in Figs. 11 and 12. In Fig. 11, there are three hotspot grids h_1, h_2, h_3 and four saver nets s_1, s_2, s_3, s_4 . Hence, in Fig. 12, there are three nodes h_1, h_2 and h_3 in H_g and s_1, s_2, s_3 , and s_4 in S_n . In Fig. 11, let the search region of hotspot grid h_1 only contain one saver net s_1 . Therefore, only one saver net assignment edge e_{h_1,s_1} is constructed. As to hotspot grids h_2 and h_3 , let their saver nets be $\{s_2, s_4\}$ and $\{s_3, s_4\}$, respectively. Hence, saver net assignment edges, $e_{h_2,s_2}, e_{h_2,s_4}, e_{h_3,s_3}$, and e_{h_3,s_4} are constructed as shown in Fig. 12.

Since there are three tiers, we have three sets of *saver_net_at_tier* nodes, T_1, T_2 , and T_3 , and two sets of signal TSVs, S_{tsv_1} and S_{tsv_2} for two intertier connections. To

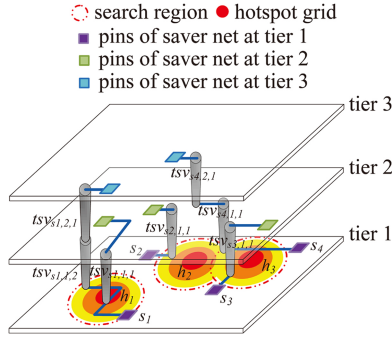


Fig. 11. Example of flow network model.

select specific TSVs of each saver net, connecting edges are constructed between nodes s_i in S_n to *saver_net_at_tier* nodes in T_1 . Then, for saver net s_1 , there are two TSVs $tsv_{s_1,1,1}$, $tsv_{s_1,1,2}$ across tier 1 for selection in Fig. 11. Therefore, node $t_{s_1,1}$ connects two nodes, $tsv_{s_1,1,1}$ and $tsv_{s_1,1,2}$ at tier 1, and TSV assignment edges $e_{t_{s_1,1}, tsv_{s_1,1,1}}$ and $e_{t_{s_1,1}, tsv_{s_1,1,2}}$ are constructed as shown in Fig. 12. Moreover, layer connecting edges $e_{tsv_{s_1,1,1}, t_{s_1,2}}$ and $e_{tsv_{s_1,1,2}, t_{s_1,2}}$ are also created whose edge cost is set to 0. For other saver nets s_2 , s_3 , and s_4 , their TSV assignment edges and layer connecting edges are to be constructed similarly. Note that saver nets s_2 and s_3 connect nodes at tier 1 and tier 2 only. Hence, there is no nodes in S_{tsv_2} . Finally, outgoing edges $e_{t_{s_1,3}, t}$, $e_{t_{s_2,2}, t}$, $e_{t_{s_3,2}, t}$, and $e_{t_{s_4,3}, t}$ are connected from the topmost *saver_net_at_tier* nodes to the destination node t .

3) *Cost of Saver Net Assignment Edges*: The objective of saver net assignment edges is to find the best saver net for each hotspot grid. Therefore, the wiring overhead incurred by TSV relocation and temperature reduction factors should be taken into consideration simultaneously. The cost function of saver net assignment edges, $EdgeCost(h_g, s_e)$, is defined as follows:

$$EdgeCost(h_g, s_e) = \left(\sum_{k=1}^n Power_{i,j,k} + \sum_{k=1}^n TSV_{i,j,k} + \sum_{k=b}^t WSDens_{i,j,k} + WL_{h_g, s_e} \right) / (StackingDegree_{s_e}) \quad (3)$$

where n , b , t are the number of tiers in the design, the bottommost and the topmost tier number of intertier net. All factors of (3) will be explained in sequence.

First, we define $Power_{i,j,k}$ to denote power density where i, j, k denote the coordinates of the hotspot grid h_g in x, y, z axis direction

$$Power_{i,j,k} = 1 - \left(\frac{PowDensity_{i,j,k}}{MaximumPowDensity} \right). \quad (4)$$

$MaximumPowDensity$ is the maximum power density in the circuit. $PowDensity_{i,j,k}$ is the power density in grid (i, j, k) . A smaller value of $Power_{i,j,k}$ represents a higher power density grid that needs more stacked signal TSVs to dissipate its heat.

Second, to consider the distribution of signal TSVs, $TSV_{i,j,k}$ is defined as follows:

$$TSV_{i,j,k} = \frac{TSVNum_{i,j,k}}{MaximumTSVNum}. \quad (5)$$

$MaximumTSVNum$ is maximum number of signal TSVs among all hotspot grids. $TSVNum_{i,j,k}$ is the number of signal TSVs in hotspot grid (i, j, k) . A small value of $TSV_{i,j,k}$ at grid (i, j, k) means less effective of the TSV to dissipate heat at grid (i, j, k) .

Next, to move signal TSVs to the same 2-D location across all tiers will change the routing topology and increase wiring overhead. Therefore, WL_{h_g, s_e} is defined as follows:

$$WL_{h_g, s_e} = \frac{WireDiff_{h_g, s_e}}{MaximumWirelength} \quad (6)$$

where $WireDiff_{h_g, s_e}$ is the routing wirelength difference which is calculated by $WireBefore_{h_g, s_e}$ denoting the routing wirelength of saver net s_e before moving subtracted from $WireAfter_{h_g, s_e}$ denoting the new routing wirelength when saver net s_e in all tiers is moved to the hotspot grid h_g . The $MaximumWirelength$ is the maximum value of $WireDiff_{h_g, s_e}$ for all saver nets s_e and hotspot grids h_g . The larger edge cost of WL_{h_g, s_e} is, the more wiring overhead saver net s_e induces.

When a stacked signal TSV is placed in the hotspot grid, and if there is not enough whitespace, overlapping cells occur. Overlapping cells need to be moved to neighboring area and thus increase wirelength. For this reason, $WSDens_{i,j,k}$ is defined to denote whitespace density for grid (i, j, k)

$$WSDens_{i,j,k} = 1 - \left(\frac{WhiteSpaceArea_{i,j,k}}{TotalArea_{i,j,k}} \right) \quad (7)$$

where $TotalArea_{i,j,k}$ and $WhiteSpaceArea_{i,j,k}$ are the total area and the whitespace area in grid (i, j, k) , respectively. A small value of $WSDens_{i,j,k}$ represents a lot of whitespace in grid (i, j, k) .

Finally, if stacked TSVs of saver net are across many tiers, the heat flow can be delivered close to the heat sink. Therefore, $StackingDegree_{s_e}$ is defined to denote stacking degree of saver net s_e

$$StackingDegree_{s_e} = t - b + 1. \quad (8)$$

A large number means the saver net s_e has good heat dissipation ability. In this cost function, these five factors have the same weight in order to consider wirelength overhead and circuit temperature simultaneously.

4) *Cost of TSV Assignment Edges*: The objective of TSV assignment edges is to select the suitable TSVs and construct stacked TSV for reducing the temperature in the circuit. Therefore, the heat dissipation ability of relocated TSVs locations in saver net should be taken into consideration. The cost function of TSV assignment edges, $EdgeCost(t_{s_e, k}, tsv_{s_e, k, q})$, is defined as follows:

$$EdgeCost(t_{s_e, k}, tsv_{s_e, k, q}) = \frac{ThermalCri_{i,j,k}}{StackingDegree_{s_e}} \quad (9)$$

where i, j, k denote the coordinates of the relocated TSV location grid in x, y, z axis direction, and $ThermalCri_{i,j,k}$ is the thermal criticality defined in (2). $StackingDegree_{s_e}$ is the stacking degree of saver net s_e defined in (8). Based on this cost function, a smaller thermal criticality value denotes the higher priority for TSV to be moved.

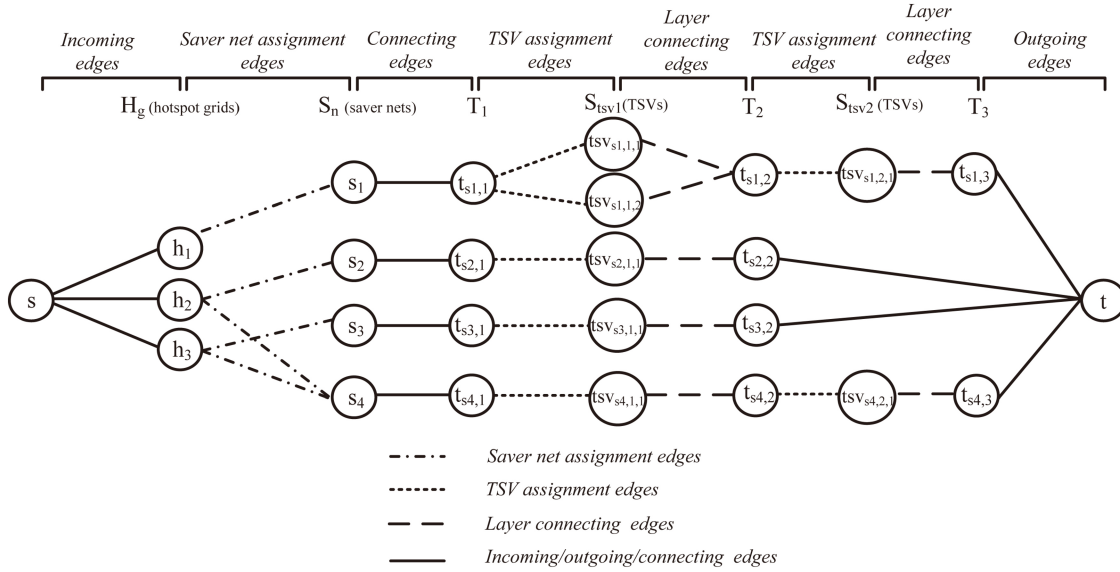


Fig. 12. Flow network for TSV relocation.

D. TSV Restoration for Timing Violation Stage

Stacked signal TSVs are relocated to high temperature regions in the circuit during the relocation step. Additional wiring overhead may be incurred. To prevent the extra wire from degrading the circuit performance, in this section, we will describe our TSV restoration for timing violation stage. Fig. 13 shows the flow. At the first, the timing analysis will be performed, and the delay value for each path will be calculated. If the path delay is greater than the maximum delay of original circuits [1], the path will be identified as timing violation path. For each timing violation path, one of its intertier nets TSVs is to be selected for restoring to its original location [1]. To minimize the total number of stacked TSVs to be restored for timing constraint, our algorithm selects the TSVs which impact more timing violation paths [1]. The restoration stage continues until no timing violation path in the circuit.

To select the minimum TSVs for all timing violation paths, we model it as a minimum set covering problem. Given an universe set U which contains all timing violation paths cp_i and a family S of subsets of U . For each subset $s_i \in S$, it represents the set of timing violation paths that TSV i resides on. Then, a cover C which is a subset of S and whose union is U corresponds to a set of TSVs whose restoration to their original positions will improve all timing violation paths. A minimized C denotes a minimum number of TSVs.

Fig. 14 demonstrates a modeling example. In Fig. 14(a), there are three timing violation paths $\{cp_1, cp_2, cp_3\} \in U$, and seven TSVs $\{s_1, s_2, s_3, s_4, s_5, s_6, s_7\} \in S$. For path cp_1 , three TSVs $\{s_1, s_2, s_4\}$ reside on it. As to paths cp_2 and cp_3 , $\{s_1, s_3, s_5\}$ and $\{s_6, s_7\}$ are on the paths, respectively. Fig. 14(b) is the covering relations between TSVs and timing violation paths. For column cp_1 , the rows s_1, s_2, s_4 are set to 1 because s_1, s_2, s_4 are on the path of cp_1 . Similarly, for columns cp_2 and cp_3 , the rows $\{s_1, s_3, s_5\}$ and $\{s_6, s_7\}$ are set to 1, respectively.

To improve timing of paths in the circuit, for each paths cp_i , one TSV must be selected for restoration. Moreover, the total number of TSVs should be minimized. Let us give an example from Fig. 14(a). In this figure, s_2 can be selected for cp_1 , s_3 for cp_2 , and s_6 for cp_3 . Therefore, a selection $\{s_2, s_3, s_6\}$ can cover all timing violation paths, and its size is 3. On the other hand, s_1 resides on both cp_1 and cp_2 . The timing of cp_1 and cp_2 can be improved if s_1 is selected for restoration. Consequently, another selection $\{s_1, s_6\}$ also covers all paths and have less number of TSVs for restoration.

As the set covering problem is a NP-complete problem, we proposed a heuristic method to find the minimum number of TSVs. The basic idea of our heuristic method is to select the TSV which resides on the largest number of uncover timing violation paths at each iteration. Let us demonstrate how to select TSVs for restoration in Fig. 14. At the beginning, all timing violation paths are uncovered, and s_1 is selected at the first iteration because it covers the most number of uncovered timing violation paths $\{cp_1, cp_2\}$. After that, cp_1 and cp_2 are identified as covered. Next, s_6 is selected at the second decision because it covers the only uncovered path $cp_3 \in U$. Finally, a cover solution $c_i = \{s_1, s_6\} = \{\{cp_1, cp_2\}, \{cp_3\}\}$ is determined. TSVs $\{s_1, s_6\}$ are restored to their original locations for timing constraint. The next iteration starts with timing analysis. The iteration continues till the timing constraint is met.

V. EXPERIMENT

A. Experimental Environment

In this section, we will discuss our experimental setup. Our benchmarking set includes one circuit *b19* from 2005 IWLS benchmarks [21] and industrial circuits (*ckt2* to *ckt5*) [1]. The selection of our benchmarking set is the same as that in [1]. Our 3-D placement results are produced by a partitioning-driven placement for 3-D ICs [1]. In [1], a new

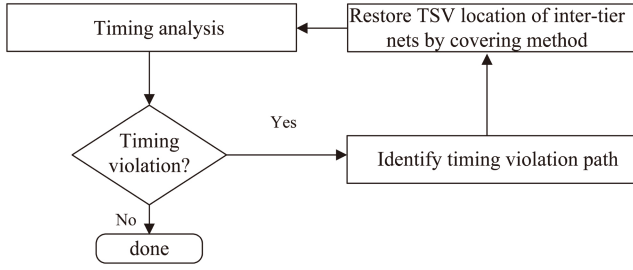


Fig. 13. Flow of TSV restoration for timing violation path.

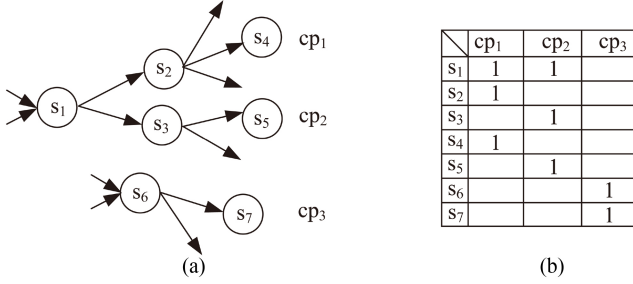


Fig. 14. TSV restoration for timing violation paths. (a) TSVs of timing violation path. (b) Set cover relation between TSV and timing violation path.

3-D physical design flow is proposed to minimize the total wirelength and signal TSV count. The results produced by [1] are fed to *FLUTE* [20] to estimate global routing wirelength. *FLUTE* is a fast package for rectilinear Steiner minimal tree (RSMT) construction by using table lookup and net reduction technique. These 3-D placement results [1] are also used as our baseline for comparison. The LEDA package [22] is used to implement the network flow algorithm.

For thermal analysis, we use the following thermal resistivity values: silicon is $6670 (\mu\text{m} \cdot \text{K/W})$, copper $2500 (\mu\text{m} \cdot \text{K/W})$, low-K material $10^6 (\mu\text{m} \cdot \text{K/W})$, and bonding material $3 \times 10^6 (\mu\text{m} \cdot \text{K/W})$. The thickness of wafer, SiO_2 in TSV, and total metal layers are shown in Table I. The power dissipation of each cell has a random distribution ranging from 1 to $1 \times 10^7 \text{ W/m}^2$ [3]. Moreover, the heat sinks are attached to the topmost layer of chips, and the ambient temperature is set to 25°C .

For timing analysis, we assume that all four dies have the same resistance and capacitance for wire and TSV where $R_{\text{wire}} = 86 (\Omega/\text{mm})$, $C_{\text{wire}} = 396 (\text{fF}/\text{mm})$, and $R_{\text{via}} = 0.172 (\Omega/\text{mm})$, $C_{\text{via}} = 1943.8 (\text{fF}/\text{mm})$. Then, the 3-D Elmore delay model is used to estimate the maximum-path delay value [3].

In this experiment, the size of signal TSV is $4.94 \mu\text{m} \times 4.94 \mu\text{m}$ [1] and the size of thermal grid $25 \mu\text{m}$ in our grid-based thermal model [2]. Moreover, all benchmarks are placed into four tiers. The flat level, timing constraint calculated by 3-D Elmore delay model [3], and the total power of each circuit are shown in Table II. Table III shows the net information of benchmarks. The CMOS BULK process with copper TSV and metal layer in 45nm technology is assumed.

B. Experimental Result

1) *Comparison with Original Circuit:* Table IV shows the comparisons of baseline and ours. The *TSV* column means the

TABLE II
CHARACTERISTICS OF BENCHMARKS

ckt	# gates	# nets	# signal TSVs	flat level #tile	timing (ns)	power(mW)
b19	169K	170K	3495	$20 \times 23 \times 4$	17.46	1.84
ckt2	330K	329K	21146	$35 \times 32 \times 4$	3.93	4.48
ckt3	660K	661K	46621	$46 \times 48 \times 4$	3.09	8.83
ckt4	741K	741K	33560	$50 \times 48 \times 4$	11.69	9.60
ckt5	905K	903K	22498	$50 \times 48 \times 4$	8.72	11.52

TABLE III
NET INFORMATION OF BENCHMARKS

ckt	# 2-tier nets	# 3-tier nets	# 4-tier nets
b19	1976	352	107
ckt2	3391	1873	1552
ckt3	8781	6868	3731
ckt4	7131	3128	1176
ckt5	9493	2241	571

total number of TSV, and the *S.TSV* column means the total number of stacked signal TSV in the circuit. The number of stacked signal TSV is defined by

$$\sum_{\substack{n_i \in \text{all nets} \\ p \in \text{all locations}}} S.TSV_{n_i, p}$$

where n_i is a net and $S.TSV_{n_i, p}$ is the number of signal TSVs at location p in net n_i if more than two consecutive TSVs are stacking at the same 2-D location p . The *wire* column means the total wirelength in the circuit. *Tmax* column represents the maximum temperature among all grids in the circuit, and *Tavg* column represents the average temperature in steady-state thermal analysis.

On all benchmarks, we can see that the number of signal TSVs that are stacked is increased substantially (295%). With the stacked signal TSV, the maximum and average temperature reductions are 17% and 13% respectively as compared with baseline results. Especially, benchmark *b19* can achieve 37°C temperature reduction. In the baseline circuit, the signal TSVs are distributed in different 2-D locations. Thus, the heat flow cannot pass through these signal TSVs effectively. On the other hand, we can see that *ckt3* has only about 6% improvement in terms of temperature reduction. The reason is that many signal TSVs of *ckt3* have already been stacked and placed close to hotspot regions in original placement result. Constructing stacked TSVs results in restricted routing topology and thus routing overhead. Our wirelength overhead of all benchmarks is about 3% compared with that of baseline.

2) *Comparison with Thermal TSV Insertion Method:* To compare our paper with a thermal TSV insertion paper by Cong and Zhang [2] where dummy TSVs are inserted to reduce peak temperature of circuit, we perform the following experiment. Since the paper by Cong and Zhang [2] requires a temperature constraint, in this experiment, the temperature obtained in our result is set as temperature constraint. The same baseline circuits are used as input circuits [1] for both thermal-TSV insertion method [2] and our method. Then, additional thermal TSVs are inserted based on power distribution and heat flow [2] in the original circuits. Table V shows the comparisons of thermal via insertion method [2]

TABLE IV
COMPARISON BETWEEN BASELINE AND OURS ON WIRELENGTH AND TEMPERATURE

ckt	baseline [1]					ours				
	<i>TSV</i>	<i>S.TSV</i>	<i>Tmax</i> (°C)	<i>Tavg</i> (°C)	wire(μm)	<i>TSV</i>	<i>S.TSV</i>	<i>Tmax</i>	<i>Tavg</i>	wire
b19	3495	101	168.6	155.7	3.46e+06	3495	489	131.0	124.4	3.58e+06
ckt2	21146	566	91.5	86.3	9.21e+06	21146	2498	84.5	81.3	9.59e+06
ckt3	46621	1215	91.4	84.6	2.53e+07	46621	5033	86.3	82.5	2.51e+07
ckt4	33560	545	110.5	99.2	2.55e+07	33560	2697	93.5	86.2	2.68e+07
ckt5	22498	1548	147.0	133.3	1.60e+07	22498	2156	124.0	116.5	1.67e+07
Ratio	1	1	1	1	1	1	2.95	0.83	0.87	1.03

TABLE V
COMPARISON BETWEEN THERMAL VIA INSERTION METHOD [2] AND OURS ON TSV COUNT AND TEMPERATURE

ckt	thermal-TSV insertion method [2]			ours		
	<i>T.TSV</i>	<i>Tmax</i> (°C)	<i>Tavg</i> (°C)	<i>T.TSV</i>	<i>Tmax</i>	<i>Tavg</i>
b19	4580	129.4	120.6	3495	131.0	124.4
ckt2	23748	84.3	81.4	21146	84.5	81.3
ckt3	53549	87.3	82.3	46621	86.3	82.5
ckt4	39562	94.9	88.2	33560	93.5	86.2
ckt5	31116	122.9	116.5	22498	124.0	116.5
Ratio	1.23	0.83	0.87	1	0.83	0.87

TABLE VI
COMPARISON BETWEEN MOVABLE TSV RELOCATION METHOD [3] AND OURS ON TEMPERATURE AND WIRELENGTH

ckt	movable TSV relocation method [3]			ours		
	<i>Tmax</i> (°C)	<i>Tavg</i> (°C)	wire(μm)	<i>Tmax</i>	<i>Tavg</i>	wire(μm)
b19	154.3	141.1	3.50e+06	131.0	124.4	3.58e+06
ckt2	85.9	82.3	9.39e+06	84.5	81.3	9.59e+06
ckt3	87.8	83.6	2.47e+07	86.3	82.5	2.51e+07
ckt4	101.4	90.7	2.58e+07	93.5	86.2	2.68e+07
ckt5	126.9	117.9	1.64e+07	124.0	116.5	1.67e+07
Ratio	0.91	0.92	1.01	0.83	0.87	1.03

TABLE VII
ROUTING CONGESTION

ckt	baseline		ours	
	<i>AvgR_d</i>	<i>MaxR_d</i>	<i>AvgR_d</i>	<i>MaxR_d</i>
b19	35.5	119.5	36.3	111.5
ckt2	37.1	138.5	38.4	126
ckt3	51.7	185	50.7	184
ckt4	49.4	236.5	51.6	215.5
ckt5	28.85	131	31.5	98.5
Ratio	1	1	1.03	0.90

and ours using the result of baseline as reference value. Column labeled *T.TSV* is the total numbers of TSVs including signal and thermal TSVs. We can see that with the same temperature constraint, the thermal TSV insertion method has in average 23% as more number of TSVs than ours. Unlike the thermal TSV insertion method, our algorithm does not require any unused whitespace to place thermal TSVs. As a result, utilizing stacked signal TSVs cannot only improve the thermal dissipation ability but also eliminate the need of whitespace for thermal TSVs. The number of stacked TSV (column *S.TSV*) and wirelength (column *wire*) by Cong and Zhang [2] are the same as the baseline results in Table IV because thermal-TSV insertion method only adds additional thermal TSVs to the whitespace without changing the position of signal TSVs. Hence, they are not shown in Table V.

3) *Comparison With Movable TSV Relocation Method:* To compare our paper with a previous paper that does not add thermal TSV, we implemented Pathak and Lim's [3] paper. Pathak and Lim [3] proposed a 3-D tree refinement algorithm with TSV relocation where signal TSVs are relocated to hotspot area under movable ranges without any performance loss. Signal TSVs are classified to movable and nonmovable TSVs. Movable TSVs are those TSVs whose movement will not incur any performance loss. The ranges of movable TSVs are calculated. Then, these movable TSVs are placed to hotspot regions under movable ranges. The same baseline circuits and timing constraint are used for both movable TSV relocation method [3] and ours. Table VI shows the temperature and wiring overhead comparison results as compared with those of baseline. In movable TSV relocation method [3], the peak and average temperature have 9% and 8% improvement, respectively, as compared with the baseline setting. Under the same number of signal TSVs, our peak and average temperature have extra 8% and 5% improvement, respectively, as compared with movable TSV relocation method [3]. Our temperature performance is better because the location of signal TSV by

movable TSV relocation method is restricted by the range and their method does not fully utilize the thermal dissipation capability of stacked TSVs. In total wirelength, our method has 2% more wiring overhead as compared with the movable TSV relocation method [3]. The reason is that our stacked TSVs are allocated to farther hotspot grids as long as the circuit timing constraint is not violated. Both the paper by Pathak and Lim [3] and ours have no performance loss.

4) *Global Routing Solution Profiling:* Extra wiring overhead may affect routing solutions in the routing step. To observe the impact in global routing stage, we have estimated congestion in the circuit. The estimation is performed the same as in Section IV-B3. At first, the Steiner tree of a net is generated by FLUTE [20] and then all two-pin nets of the tree are routed by pattern route. For each edge of a global routing model, its routing demand is computed. Table VII shows the estimation result. Column *MaxR_d* means the maximum routing demand of an edge while column *AvgR_d* means the average routing demand. According to the result, the average routing demand of edge increases 3% which indeed corresponds to overall wirelength overhead. Our initial TSV assignment stage considers the routing congestion issue and avoids assigning TSVs to congested grids. Consequently, the maximum routing demand of edge reduces about 10% compared with the original baseline.

TABLE VIII
COMPARISON RESULTS OF OUR APPROACH WITH AND WITHOUT TSV RESTORATION STAGE

ckt	Without TSV restoration stage					With TSV restoration stage				
	S.TSV	$T_{max}(^{\circ}C)$	$T_{avg}(^{\circ}C)$	wire (μm)	$Delay_{ratio}$	S.TSV	T_{max}	T_{avg}	wire (μm)	
b19	501	129.1	122.0	3.59e+06	1.01	489	131.0	124.4	3.58e+06	
ckt2	2513	84.3	81.5	9.65e+06	1.02	2498	84.5	81.3	9.59e+06	
ckt3	5068	86.0	82.5	2.51e+07	1.07	5033	86.3	82.5	2.51e+07	
ckt4	2723	93.4	86.0	2.70e+07	1.06	2697	93.5	86.2	2.68e+07	
ckt5	2163	123.0	116.2	1.68e+07	1.02	2156	124.0	116.5	1.67e+07	
Ratio	1	1	1	1	1.04	0.99	1.01	1	0.99	

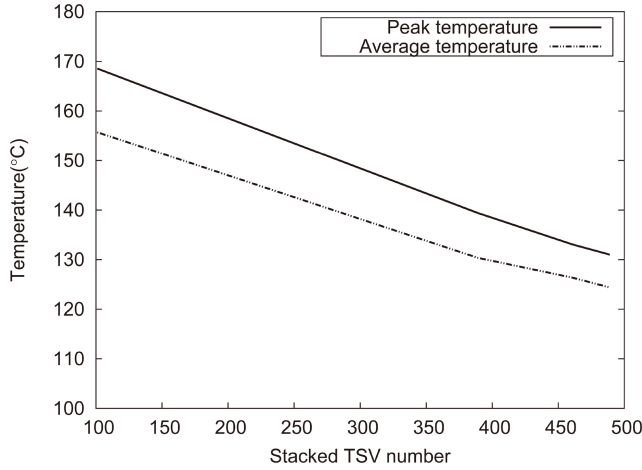


Fig. 15. Relationship between the number of stacked signal TSV and temperature in *b19*.

5) *Temperature Versus Stacked Signal TSV Number*: To understand the thermal impact of stacked signal TSV, we draw the relationship between the number of stacked signal TSVs and temperature in *b19* shown in Fig. 15. We can see that both the peak and the average temperature are decreased while the number of stacked signal TSV is increased. It also explains why stacked signal TSVs are effective in 3-D IC thermal dissipation.

6) *Performance Analysis of Each Step*: In our three-stage algorithm, the objective of stacked-TSV assignment and relocation is to minimize the chip temperature and wiring overhead. To observe individual effect, we examine each step in Fig. 16. The vertical axis is the percentage while the horizontal axis represents the temperature reduction and wirelength overhead. In the peak temperature, we can see that simply executing initial TSV assignment step has about 9% reduction while stacked-TSV relocation step has nearly 14%. In average temperature, the trend is similar. By combining these two techniques, our algorithm can have 17% improvement of the temperature reduction. To observe the effectiveness of TSV restoration stage, Table VIII shows a comparison results of our approach with and without TSV restoration stage. The $Delay_{ratio}$ column represents the circuit timing normalized to the performance constraint. Although the stacked TSVs are restored for performance constraint, our algorithm restores less than 1% TSV and produces competitive temperature results as compared with the method without TSV restoration stage. Moreover, a notable result of our TSV restoration stage is that it eliminates 4% timing degradation.

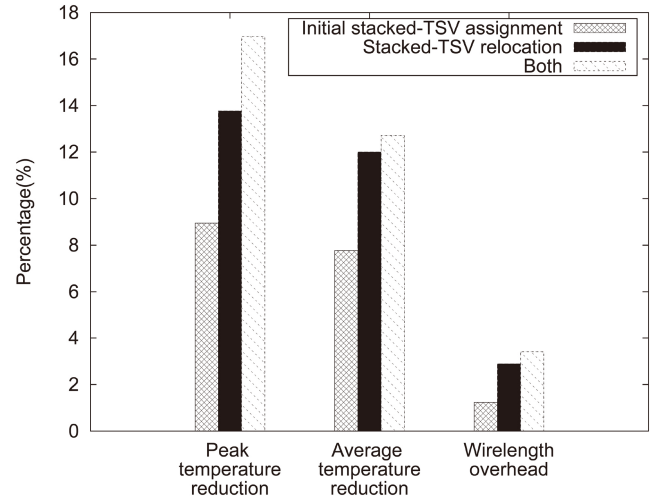


Fig. 16. Analysis of each step.

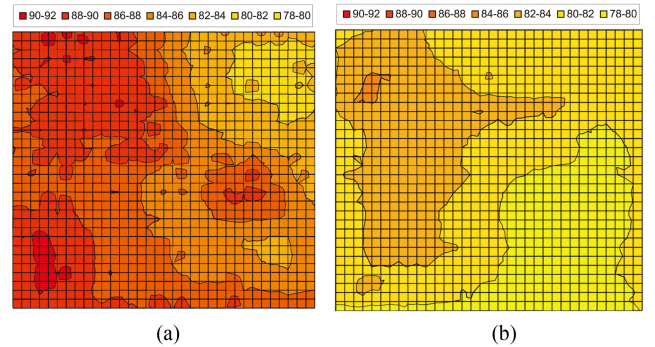


Fig. 17. Isothermal comparison in ckt2. (a) Original. (b) Ours.

7) *Thermal Map of One Testcase*: Finally, Fig. 17 shows the isothermal distribution comparisons between baseline and ours in ckt2. This figure also validates the aforementioned thermal analysis. Our algorithm can solve the hotspot regions and improve the ability of thermal dissipation.

VI. CONCLUSION

A new integrated design methodology, i.e., stacked signal TSV, was developed to dissipate heat. Based on this structure, a three-stage TSV locating algorithm has been proposed to construct the stacked signal TSVs. The superior thermal conductance of TSV is fully utilized by our method to optimize the chip temperature. The experimental results show that our method is effective in minimizing temperature by 17% with

small wiring overhead and no performance degradation. Compared with thermal-TSV insertion method [2], our proposed algorithm has zero hardware overhead incurred by thermal-TSV. Compared with movable TSV relocation method [3] that does not add thermal TSVs, our algorithm has 8% more peak temperature reduction under the same number of signal TSVs.

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